

C Programming Lab

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Program 1: Convert Time in Seconds to Days, Hours, Minutes and Seconds

1.1 Aim

To write a C program that accepts a duration of time in seconds as input and converts it into an equivalent format expressed in days, hours, minutes and seconds.

1.2 Problem Statement

Write a C program to input time in seconds and convert it into days, hours, minutes, and seconds format.

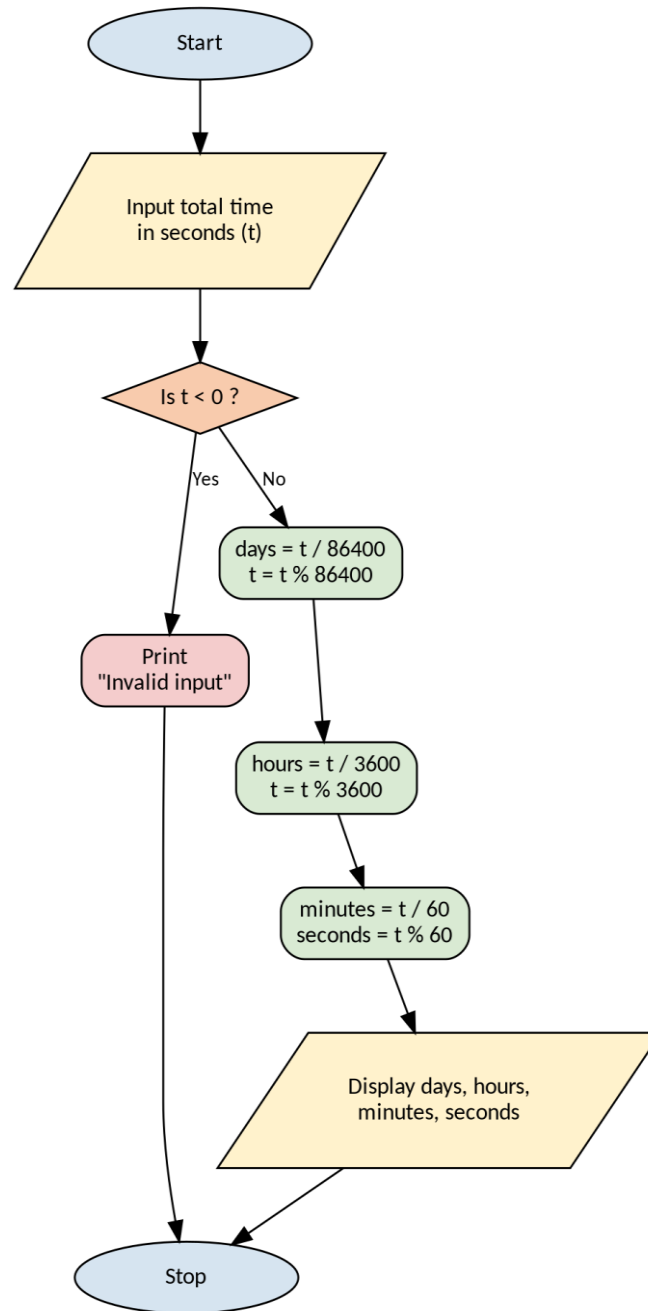
1.3 Algorithm

1. Start the program.
2. Declare variables totalSeconds, days, hours, minutes, seconds and remaining.
3. Read the value of totalSeconds from the user.
4. If totalSeconds is negative, display an "Invalid input" message and stop.
5. Compute $\text{days} = \text{totalSeconds} / 86400$ and update $\text{remaining} = \text{totalSeconds} \% 86400$.
6. Compute $\text{hours} = \text{remaining} / 3600$ and update $\text{remaining} = \text{remaining} \% 3600$.
7. Compute $\text{minutes} = \text{remaining} / 60$ and $\text{seconds} = \text{remaining} \% 60$.
8. Display the computed days, hours, minutes and seconds.
9. Stop the program.

1.4 Pseudocode

```
BEGIN
  READ totalSeconds
  IF totalSeconds < 0 THEN
    PRINT "Invalid input"
  ELSE
    days      <- totalSeconds / 86400
    remaining <- totalSeconds MOD 86400
    hours     <- remaining / 3600
    remaining <- remaining MOD 3600
    minutes   <- remaining / 60
    seconds   <- remaining MOD 60
    PRINT days, hours, minutes, seconds
  END IF
END
```

1.5 Flowchart



1.6 Test Case Table

Test No.	Input (seconds)	Expected Output	Actual Output	Result
1	90061	1d 1h 1m 1s	1 days, 1 hours, 1 minutes, 1 seconds	Pass
2	3661	0d 1h 1m 1s	0 days, 1 hours, 1 minutes, 1 seconds	Pass
3	59	0d 0h 0m 59s	0 days, 0 hours, 0 minutes, 59 seconds	Pass
4	86400	1d 0h 0m 0s	1 days, 0 hours, 0 minutes, 0 seconds	Pass

5	172800	2d 0h 0m 0s	2 days, 0 hours, 0 minutes, 0 seconds	Pass
6	-50	Invalid input	Invalid input! Time in seconds cannot be negative.	Pass

1.7 Source Code (C Program)

```
#include <stdio.h>

int main() {
    long totalSeconds, days, hours, minutes, seconds, remaining;

    printf("Enter time in seconds: ");
    scanf("%ld", &totalSeconds);

    if (totalSeconds < 0) {
        printf("Invalid input! Time in seconds cannot be negative.\n");
        return 0;
    }

    days = totalSeconds / 86400;
    remaining = totalSeconds % 86400;

    hours = remaining / 3600;
    remaining = remaining % 3600;

    minutes = remaining / 60;
    seconds = remaining % 60;

    printf("\n%ld seconds = %ld days, %ld hours, %ld minutes, %ld seconds\n",
        totalSeconds, days, hours, minutes, seconds);

    return 0;
}
```

1.8 Compilation and Execution

The program was compiled and executed successfully in the VS Code integrated terminal using the GCC compiler. The commands used and sample runs are shown below:

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\Rudra Pratap Singh> gcc time_convert.c -o time_convert.exe
PS C:\Users\Rudra Pratap Singh> .\time_convert.exe
Enter time in seconds: 90061

90061 seconds = 1 days, 1 hours, 1 minutes, 1 seconds
PS C:\Users\Rudra Pratap Singh> .\time_convert.exe
Enter time in seconds: 3661

3661 seconds = 0 days, 1 hours, 1 minutes, 1 seconds
PS C:\Users\Rudra Pratap Singh> .\time_convert.exe
Enter time in seconds: 86400

86400 seconds = 1 days, 0 hours, 0 minutes, 0 seconds
PS C:\Users\Rudra Pratap Singh> █
```

1.9 Result

The C program was executed successfully and the given time in seconds was correctly converted into days, hours, minutes and seconds for all the test cases listed above. The program's output matched the expected output in every case.

Program 2: Check Whether a Given Year is a Leap Year

2.1 Aim

To write a C program that accepts a year as input and determines whether it is a leap year or not.

2.2 Problem Statement

Write a program to input a year and check whether it is a leap year or not.

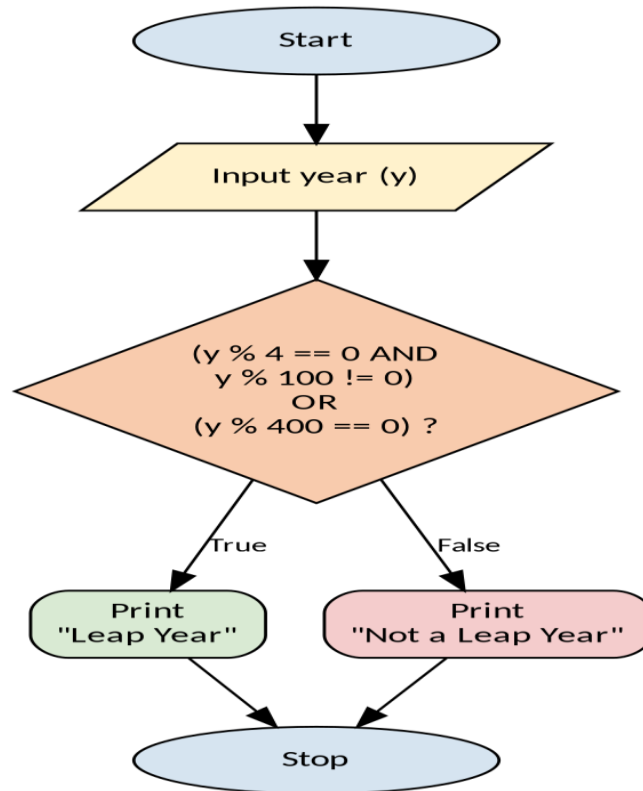
2.3 Algorithm

1. Start the program.
2. Declare an integer variable year.
3. Read the value of year from the user.
4. Check if (year is divisible by 4 AND not divisible by 100) OR (year is divisible by 400).
5. If the condition is true, display that the year is a leap year.
6. Otherwise, display that the year is not a leap year.
7. Stop the program.

2.4 Pseudocode

```
BEGIN
  READ year
  IF (year MOD 4 == 0 AND year MOD 100 != 0) OR (year MOD 400 == 0) THEN
    PRINT year, "is a Leap Year"
  ELSE
    PRINT year, "is Not a Leap Year"
  END IF
END
```

2.5 Flowchart



2.6 Test Case Table

Test No.	Input (year)	Expected Output	Actual Output	Result
1	2024	Leap Year	2024 is a Leap Year.	Pass
2	2023	Not a Leap Year	2023 is Not a Leap Year.	Pass
3	1900	Not a Leap Year	1900 is Not a Leap Year.	Pass
4	2000	Leap Year	2000 is a Leap Year.	Pass
5	2400	Leap Year	2400 is a Leap Year.	Pass
6	1996	Leap Year	1996 is a Leap Year.	Pass

2.7 Source Code (C Program)

```
#include <stdio.h>

int main() {
    int year;

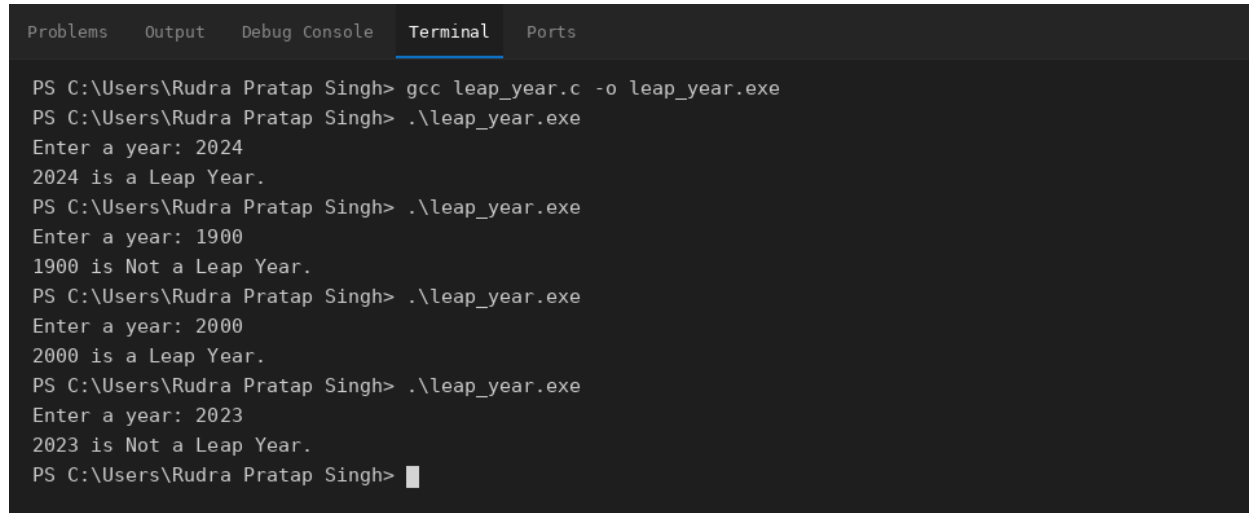
    printf("Enter a year: ");
    scanf("%d", &year);

    if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
        printf("%d is a Leap Year.\n", year);
    } else {
        printf("%d is Not a Leap Year.\n", year);
    }
}
```

```
    return 0;  
}
```

2.8 Compilation and Execution

The program was compiled and executed successfully in the VS Code integrated terminal using the GCC compiler. The commands used and sample runs are shown below:



```
Problems  Output  Debug Console  Terminal  Ports  
PS C:\Users\Rudra Pratap Singh> gcc leap_year.c -o leap_year.exe  
PS C:\Users\Rudra Pratap Singh> .\leap_year.exe  
Enter a year: 2024  
2024 is a Leap Year.  
PS C:\Users\Rudra Pratap Singh> .\leap_year.exe  
Enter a year: 1900  
1900 is Not a Leap Year.  
PS C:\Users\Rudra Pratap Singh> .\leap_year.exe  
Enter a year: 2000  
2000 is a Leap Year.  
PS C:\Users\Rudra Pratap Singh> .\leap_year.exe  
Enter a year: 2023  
2023 is Not a Leap Year.  
PS C:\Users\Rudra Pratap Singh> █
```

2.9 Result

The C program was executed successfully and correctly identified leap years and non-leap years for all the test cases, including century years (1900, 2000, 2400) that specifically test the divisible-by-100 and divisible-by-400 rules.

(Note- Theory is by AI, to be exact)